

LSTM Temporal Experiments, Findings & Next Steps

What I Did This Week

Starting from Daniel's v9 BiLSTM+Attention baseline (test $R^2 = 0.747$, val $R^2 = 0.845$), I ran three experiments to understand the val→test gap.

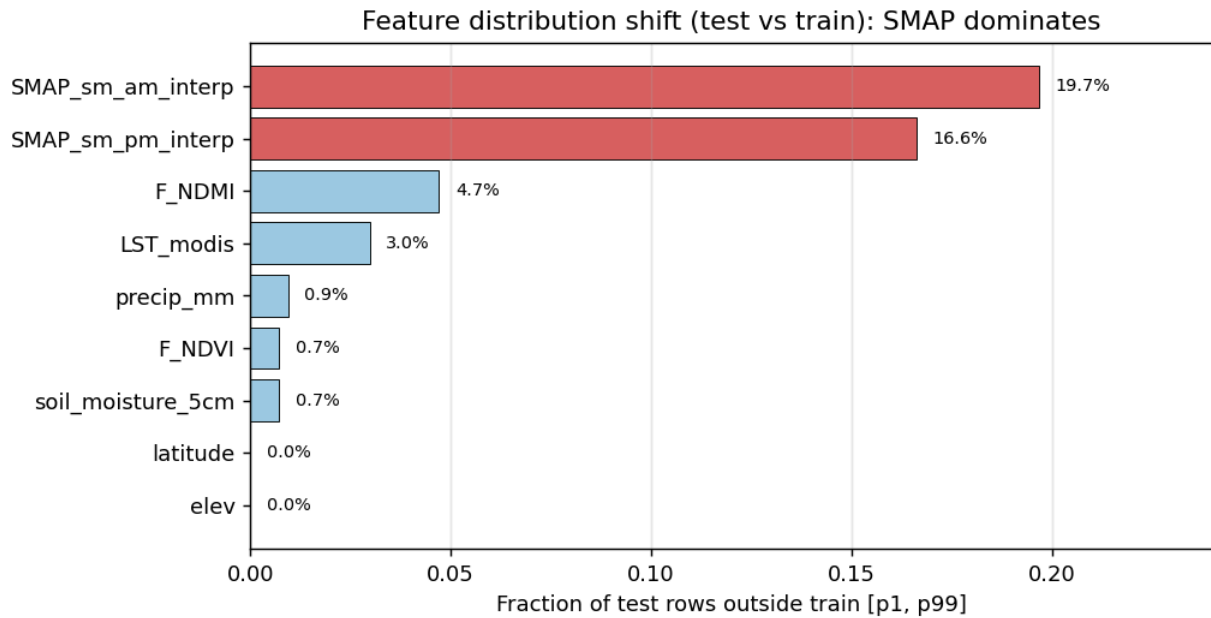
Experiment 1, Split Audit (`audit_split.py`)

Finding: the gap is temporal, not spatial.

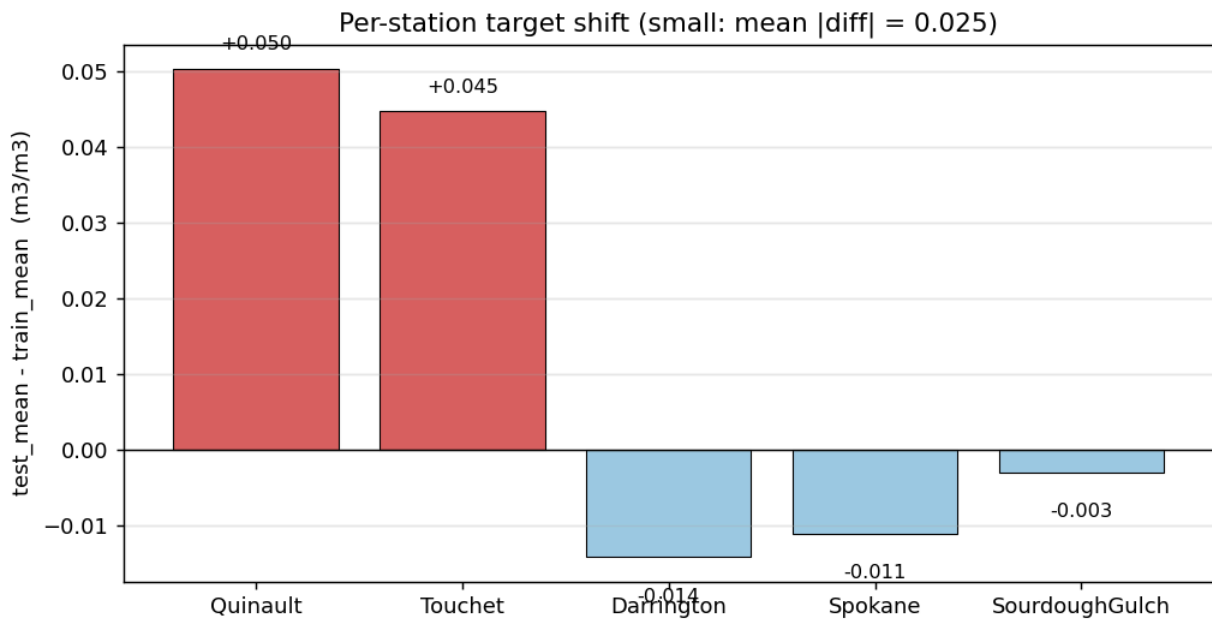
All 5 stations (Darrington, Quinault, SourdoughGulch, Spokane, Touchet) appear in both train and test (`train ∩ test = 5/5`; Touchet is absent from val). So the train→test degradation is not caused by unseen stations, it's distributional shift across years within the same stations.

The dominant shift is in SMAP features, fraction of test rows outside the train [p1, p99] range:

Feature	OOD fraction
SMAP_sm_am_interp	19.7%
SMAP_sm_pm_interp	16.6%
F_NDMI	4.7%
All other features	< 3%



Test years (2023–2025) run systematically higher SMAP soil-moisture readings than train years (2017–2020). Per-station target shift is small (mean $|\text{diff}| = 0.025 \text{ m}^3/\text{m}^3$), so the problem is in the inputs, not the target.



Experiment 2, Per-Station Target Normalization (train_v17.py)

Finding: did not help. Test $R^2 = 0.713$ vs. v9's 0.747.

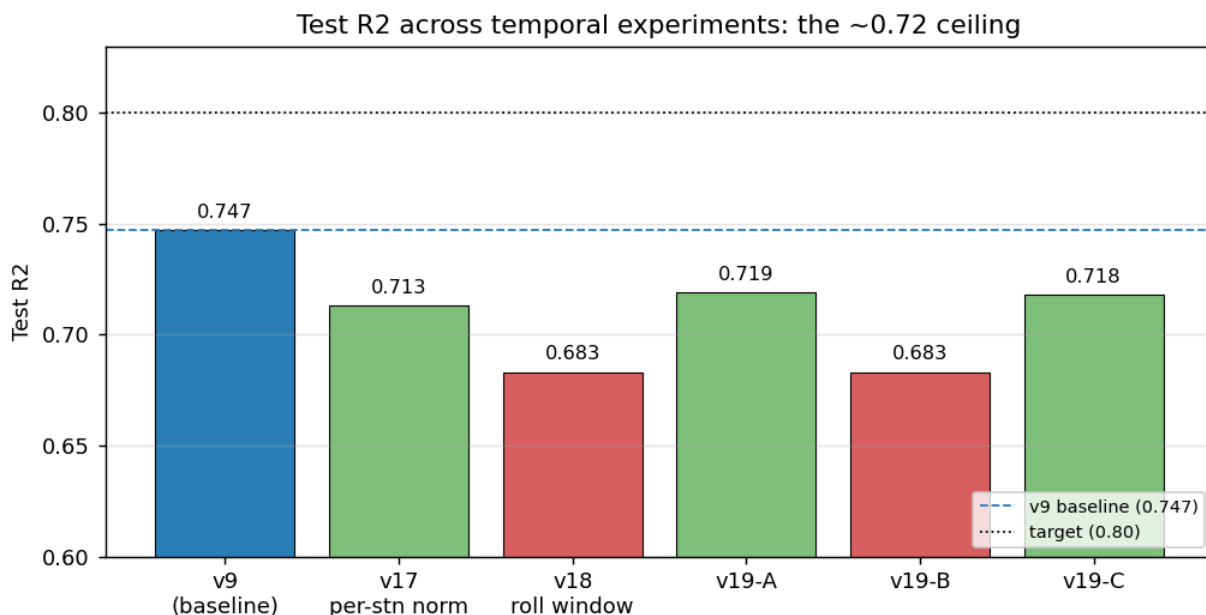
Subtracting each station's train-period target mean (train on residuals, reconstruct at inference) improved bias substantially (-0.003 vs. -0.010) but hurt variance capture. Because station means are stable across years (~ 0.025 shift), de-meaning removed useful signal without reducing the actual thing the model struggles with (input-side variance).

Experiment 3, Rolling Window & Temporal CV (`train_v18.py`, `train_v19.py`)

Finding: expanding the training window did not beat baseline on the 2023–2025 holdout.

Version	Train years	Test years	Test R^2
v9 (baseline)	2017–2020	2023–2025	0.747
v18	2017–2021	2023–2025	0.683
v19-A	2017–2020	2022–2025	0.719
v19-B	2017–2021	2023–2025	0.683
v19-C	2017–2022	2024–2025	0.718

v19 Fold B reproduces v18 exactly, which cross-validates both scripts. Every temporal configuration lands in 0.68–0.72.



Note: this result is confounded. v18 and all v19 folds use a 1-year val set with the 250-epoch OneCycleLR schedule, so early stopping fires at epoch ~ 13 , the model never gets a fair run. This means v18's drop below baseline may be a training-config artifact, not proof that more

data fails. Treat the ~ 0.72 ceiling as suggestive, not proven, until the scheduler is fixed (see next step #1).

Touchet's test set is $n=170$ with std collapsed to 0.030, its per-station R^2 is effectively a metric artifact (low-variance denominator) and slightly depresses the aggregate.

What We've Ruled Out (so far)

Approach	Result	Why
Per-station normalization (v17)	0.713	Stable means, removes useful signal
Expanded training window (v18)	0.683	Below baseline, but confounded by early stopping
Forward-chain CV (v19)	0.683– 0.719	Consistent ~ 0.72 ceiling, same confound
Architecture search (v7–v16)	0.685– 0.747	Ceiling holds across 10 variants

Possible Causes

A substantial minority ($\sim 17\text{--}20\%$) of test-year SMAP readings fall outside the range the model was trained on. Reordering which years sit in training doesn't fix this, those SMAP values simply aren't well-represented in the 5-station derived_8.0 dataset. Temporal expansion within the same stations appears to hit a wall (pending the scheduler fix above).

Next Steps

1. Fix early stopping for short/1-year-val runs (prerequisite, do first). Reconfigure OneCycleLR for the real training length or switch to ReduceLROnPlateau, then re-run v18/v19. Without this we can't trust the "more data doesn't help" conclusion that motivates step #2.
2. Add geographically diverse stations to training (highest-leverage, attacks root cause). The MoE handoff shows OR Riley (R^2 0.614), ID Murphy (0.608), CA Redding (0.497) respond reasonably to the existing model. Stations from other climate zones widen the SMAP input distribution and should better overlap the 2023–2025 range. derived_9.0 (30 stations) is already built, train an LSTM on it as the next experiment.

3. SMAP-specific input normalization. Since SMAP is the dominant OOD feature, apply a rolling per-station standardization (normalize each SMAP reading against a trailing window of that station's own history) instead of a single global train-set scaler, reduces absolute shift even when year-to-year climatology differs.
4. Binary MoE path (handoff §5). Split Dry vs. Non-Dry with the 80%-accurate gating classifier, the temporal LSTM serves as the Non-Dry expert, shrinking the target range it must cover.